



One extra line on a VMS — “USE RUSSELL ST” — cut Lonsdale St delay by 46%* bus riders –41%

PROBLEM STATEMENT

- Councils and contractors can’t foresee how a closure — alone or **overlapping other works** — hits traffic, trams and pedestrians.
- Sign effects are guessed: in London, real diversions were **one-fifth** of what drivers said they would do (Chatterjee 2002).

SOLUTION OVERVIEW

- 1 Close a lane on the **real Melbourne CBD** network; add barriers, signs and the exact **VMS wording**.
- 2 Four driver types — commuter, local, tourist, delivery — **read each sign** (DeepSeek LLM connected, with rule fallback).
- 3 A deterministic engine computes **detours, queues and delay** for drivers, riders and pedestrians, then re-tests each fix.

AI interprets — the engine calculates.

WHO USES IT & PAYS

- Contractors & councils: impact evidence for each permit.
- RPM Hire: every plan is priced from its inventory — what each extra VMS or barrier buys → the hire quote.

Before · generic warning

ROADWORK
AHEAD

FRAME 2
(OPTIONAL)

918 m queue

10,493 veh·min delay

18,693 rider·min

After · names the detour

ROADWORK
AHEAD

USE
RUSSELL ST

538 m queue

5,629 veh·min delay

10,991 rider·min

LLM-run results, 29 Sep — Lonsdale St westbound, 1 lane closed, 8 am, real CBD traffic counts

WHAT'S DIFFERENT

- RPM's VMS Preview shows how a sign **looks**; we show its **effect**.
- Mooven forecasts from past delays; we test the **wording** before deployment.
- one.network flags clashes; we count the **minutes they add**.

* Single run (1 lane closed, 8 am); DeepSeek LLM sign reading; assumed traveller mix and bus loads (25 per trip).

Tech & Progress

Uncapped

Real data, a working engine and a live demo — built since 29 Sep, 09:30

RippleTwin · FEIT Hackathon 2026

TECH STACK

DATA

Python + OSMnx → static JSON, all from open data

ENGINE

JavaScript in the browser + Canvas map; deterministic, under 10 ms per scenario

SIGN READING

DeepSeek LLM connected; cached answers + rule fallback

HOSTING

Cloudflare Workers, one URL; API keys stay server-side

OPEN DATA

OpenStreetMap · DataVic SCATS & PTV GTFS · City of Melbourne

BUILT SO FAR

Live

public demo on Cloudflare since 29 Sep

1,513

CBD road links with 8 weeks of traffic-signal counts (SCATS)

−46%

Lonsdale St driver delay when the VMS adds “USE RUSSELL ST” (1 lane, 8 am)*

−41%

delay for riders on the 15 bus routes through the closure, same sign*

+176 m

walk and 2 extra crossings for 185 people/h if one footpath closes*

3 plans

priced from RPM Hire’s inventory, A\$515–1,765; the guided VMS plan cuts the queue 918 → 79 m; 937 veh·min*

Try it: [demo link](#) + QR — team to confirm]

WORKING NOW

WORKING

Real CBD clashes: approx +33,012 veh·min; stagger by 3 days → 0 (rule-reading run)

WORKING

Compare the 3 plans side by side for drivers, riders and pedestrians

WORKING

Pick a plan and export a one-page pack: equipment list, hire quote, VMS schedule, who to notify

WORKING

DeepSeek sign reading and AI plan interpretation

1 OCT 10:00

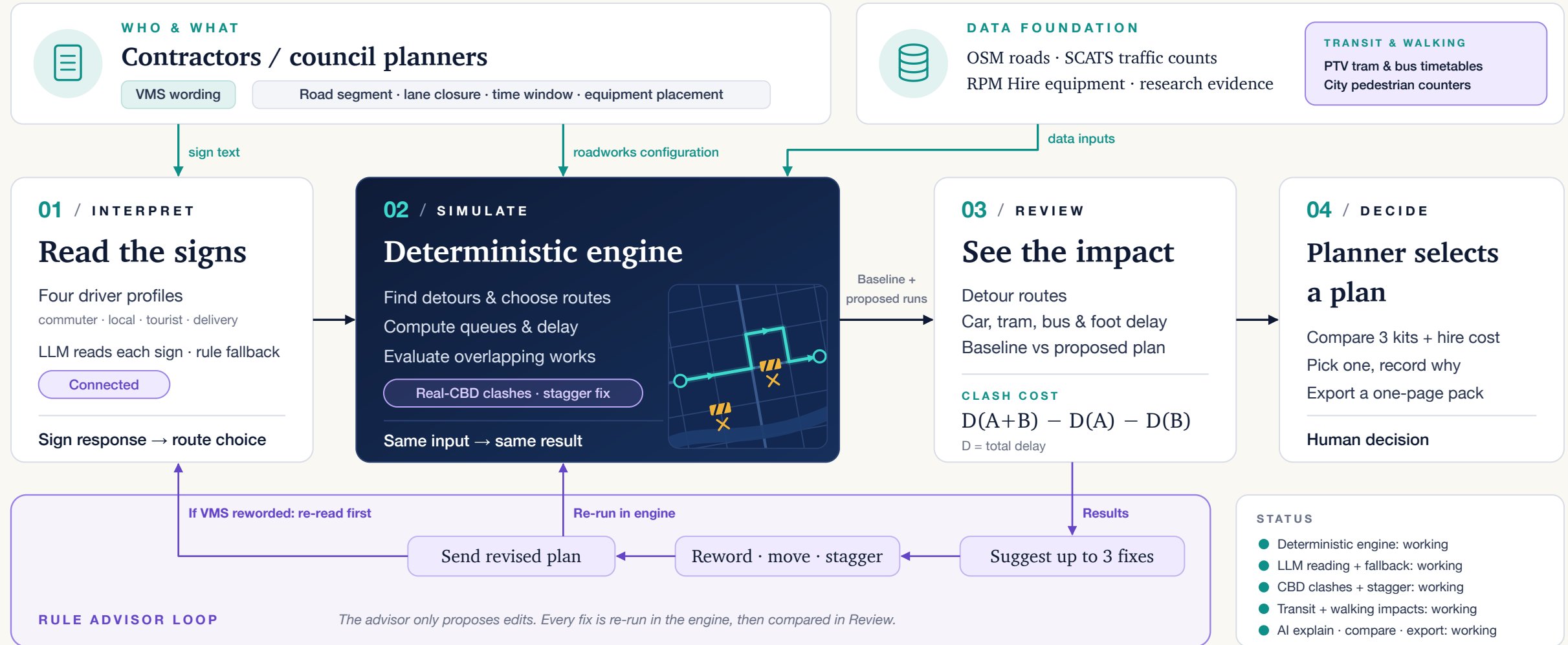
Code freeze, tagged release, public repo

* LLM sign reading, single runs; 29 of 37 behaviour parameters (incl. traveller mix) assumed or low-confidence, flagged in the app; most road and footpath flows estimated from nearby counts; bus loads, stock and hire rates assumed. Data © OpenStreetMap contributors (ODbL); DataVic (CC BY 4.0); City of Melbourne (CC BY). Status: 29 Sep 2026.

RippleTwin

BUSINESS WORKFLOW + SYSTEM ARCHITECTURE

From a roadworks plan to a tested decision



AI interprets — the engine calculates